

Dependency Parsing

Feb 2, 2026

Outline

- 1 Syntactic structure
- 2 Dependency grammar
- 3 Dependency parsing
- 4 Evaluation and progresses
- 5 Preview

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Understanding linguistic structure

- So far we have focused on how NLP deals with word meanings.
- But language goes beyond individual words.
- **Linguistic structure** is equally important for capturing how words combine together to create meaning.

Understanding linguistic structure

A **grammar** is the system of rules that defines how linguistic structures are formed and how words relate to each other within a sentence.

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- 1 Part of Speech (POS)
- 2 Dependency grammar

1. Part-of-Speech (POS)

- A word's POS determines how words fit into a sentence.
- POS is also called lexical category.
- Main POS categories (in English):
 - **N**____: reindeer, game, government
 - **V**____: play, run, believe
 - **A**____: fun, beautiful
 - **A**____: well, heavily
 - **P**____: on, into
 - **A**____/**D**____: a, the, some
 - **C**____: and, or

How do we identify POS?

■ Examples:

- *This car is very interesting.*
- *This car **mooked** fast.*
- *This **nony** car **mooked** fast.*

■ We usually identify POS by:

- **Morphology:** how a word changes form (e.g., verbs mark tense: *play* → *played*, sometimes irregularly: *go* → *went*)
- **Distribution:** where a word appears in a sentence (e.g., nouns after articles, verbs after subjects)

2. From words to phrases

- Words combine into constituents based on POS:
 - **the reindeer** = article + noun = noun phrase
 - **play games** = verb + noun phrase = verb phrase
- Constituents combine based on **phrasal category**:
 - **Noun Phrase + Verb Phrase** = Sentence

Structure over meaning

- Chomsky (1957): *“Colorless green ideas sleep furiously”*
- Nonsensical meaning, but:
 - Correct lexical and phrasal categories
 - Grammatically well-formed
- Syntax is about **structure**, not always meaning.

Understanding linguistic structure 1: Constituency

- **Constituency grammar**
- A linguistic theory that analyzes sentences as nested constituents (e.g., noun phrases, verb phrases).
- Also known as phrase structure grammar

Lexicon:

N → *reindeer, dragon, lunch, game, evening, morning*

V(trans) → *play, eat*

V(intrans) → *run, swim, dance*

Adj → *fun, beautiful, interesting*

Det → *the, a, some, many*

P → *for, in, to, at*

Phrase structure rules:

S → NP VP

VP → V(trans) NP

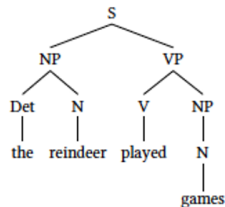
VP → V(intrans)

NP → Det (A*) N

NP → N

NP → NP PP

PP → P NP



Frameworks for analyzing grammar

- Linguists formalize sentence structure using grammar frameworks:
 - Phrase Structure Grammar

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- Linguists formalize sentence structure using grammar frameworks:
 - Phrase Structure Grammar (linguistics)
 - **Dependency Grammar (widely used in NLP)**

Understanding linguistic structure 2: Dependency

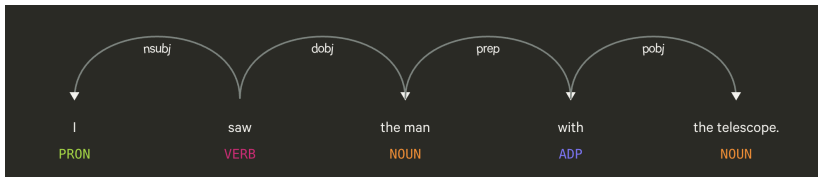
- Dependency grammar shows that syntactic structure consists of **relationships** between lexical items, normally binary asymmetric relations (“arrows”) called **dependencies**.
- Dependency structure shows which words depend on (modify, attach to, or are arguments of) which other words.

Why do we need dependency structure?

e.g., *I saw the man with the telescope.*

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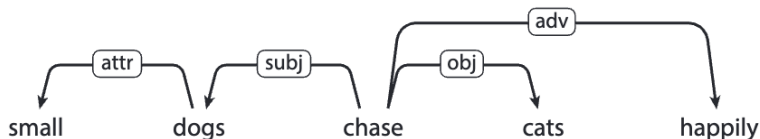
- Readers/Listeners/NLP models need to work out what modifies (attaches to) what.

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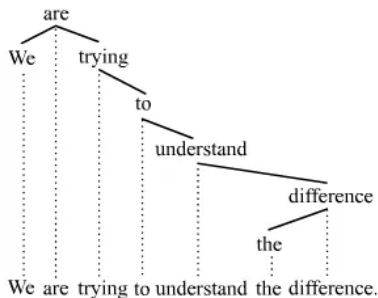
Dependency grammar and dependency structure

Dependency grammar shows that syntactic structure (of a sentence) consists of relations between lexical items, normally binary asymmetric relations (“arrows”) called dependencies.

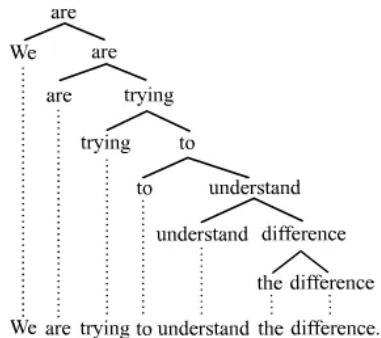


Sourced from: De Marneffe, M. C., & Nivre, J. (2019). *Dependency grammar*. *Annual Review of Linguistics*, 5(1), 197–218. Figure 1.

Dependency grammar vs. Constituency parsing



Dependency



Constituency (BPS)

The arise of annotated data & Universal Dependencies tree

What is a **treebank?** An annotated corpus that includes syntactic or morphological structure, often in the form of parse trees.

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What is a **treebank?** An annotated corpus that includes syntactic or morphological structure, often in the form of parse trees.

Milestones in treebank development:

- *Brown corpus* (1967): First general-purpose corpus; part-of-speech (PoS) tagged in 1979
- *Lancaster-IBM treebank* (Late 1980s): One of the first syntactically annotated corpora
- *The Penn treebank* (Marcus et al., 1993): Influential constituency-based treebank for English
- **Universal Dependencies (UD)**: A multilingual, cross-linguistically consistent treebank project using dependency grammar

Hakyung Sung
PSYC681 (Spring 26)

Presentation 1

Matthew & Valbona: de Marneffe et al. (2021). Universal Dependencies.

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Sources of information for dependency parsing

How do we build a **parser**, once we get the dependency information?

Methods of dependency parsing

There are several ways (including dynamic programming, graph algorithms, etc.) but we'll focus on the one method.

Presentation 2

Duvarakanath: Chen & Manning (2014). A Fast and Accurate Neural Dependency Parser.

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Gold Standard: Hand-annotated syntactic structure used for evaluating parser output.

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Metrics: (1) **UAS (Unlabeled Attachment Score)**: Correct head only; (2) **LAS (Labeled Attachment Score)**: Correct head and label

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Example:

Word	Gold Head	Gold Label	Pred Head	Pred Label
She	2	nsubj	2	nsubj
likes	0	root	0	root
chocolate	2	obj	2	nmod
very	4	advmod	4	advmod
much	2	advmod	4	advmod

Example:

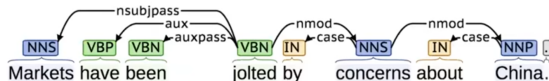
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Evaluation:

- Total dependencies: 5
- Correct heads (UAS): 4 $\rightarrow$ $\text{UAS} = 4/5 = 80\%$
- Correct heads + labels (LAS): 3 $\rightarrow$ $\text{LAS} = 3/5 = 60\%$

Dependency parsing for sentence structure

- Chen and Manning (2014) showed that neural networks can accurately determine the structure of sentences, supporting meaning interpretation.



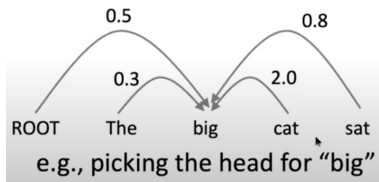
- It was the first simple, successful neural dependency parser.
- The dense representations (and non-linear classifier) let it outperform other greedy parsers in both accuracy and speed.
- This work was further developed and improved by others.

Further developments

This work was further developed and improved by others, including in particular at Google.

Graph-based dependency parsers

- Compute a score for every possible dependency (choice of head) for each word
 - Doing this well requires more than just knowing two words
 - We need good “contextual” representations of each word token
- Repeat the same process for each other word; find the best parse



A neural graph-based dependency parser

- Dozat and Manning (2017); Dozat, Qi, and Manning (2017) - This paper revived interest in graph-based dependency parsing in a neural world
 - Designed a new scoring model (i.e., biaffine) for neural dependency parsing
- Really great results!



Method	UAS	LAS (PTB WSJ SD 3.3)
Chen & Manning 2014	92.0	89.7
Weiss et al. 2015	93.99	92.05
Andor et al. 2016	94.61	92.79
Dozat & Manning 2017	95.74	94.08

- But, slower than the simple neural transition-based parsers.

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Preview

- Thursday (2/5): Lab 4 - Work with a small dataset to train dependency parser (No in-person class; **Check mycourses on time - there'll be a link for a guided video**)
- Friday (2/6): Submission deadline for **Identifying team members**
- Tuesday (2/10) RNNs, LSTMs; Presenter: Shubh Sudan